CCSA101-LAB01

Check Point Lab Setup and Administration

Consolidated Practical Laboratory Report: Parts A, B and C

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Management platform	Check Point Gaia R82 (build 779) and SmartConsole R82

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Executive summary

This report documents CCSA101-LAB01. The aim was to build a private management lab around the supplied A-SMS virtual machine, explain how Check Point management and enforcement are separated, and explore Gaia and SmartConsole without changing the supplied environment.

Environment. VMware Workstation Pro 26.0.1 hosted a single A-SMS virtual machine (8 GB RAM, 8 processors, 500 GB disk) on a VMXNET3 adapter attached to VMnet8, a NAT network on 10.1.1.0/24 with gateway 10.1.1.250 and a DHCP pool of 10.1.1.210 to 10.1.1.240. The Windows host adapter was set to 10.1.1.201/24 and A-SMS eth0 to 10.1.1.101/24. SmartConsole R82 was installed on the Windows host.

Work completed. The screenshots show eth0 up with the expected address, a 4/4 ping reply, the Gaia Portal login, and a SmartConsole login to 10.1.1.101 after the ICA fingerprint on the VMware console matched the one presented by SmartConsole. Part B answers the five theory questions. Part C records Gaia Portal, Clish, Expert mode and SmartConsole observations, all read-only.

Principal findings. A-SMS reports Gaia R82 build 779, licence status OK and no alerts. The other preconfigured devices (A-GW-Cluster, A-GW-01, A-GW-02 and the log server A-SME) show a red status, consistent with only A-SMS being deployed, and the SmartConsole log search warns that the log server A-SME is disconnected. The static default route points to 10.1.1.1 rather than the NAT gateway 10.1.1.250, NTP is not synchronised, and two of the four licence entries show an expiry date of 30/09/2026. All were recorded and left unchanged. Internet access from the VM was not tested. Limitations are listed in the Conclusion.

Part A: Setup and Management Access

A1 Software versions and sources

Item	Version observed	Source
VMware Workstation Pro	26H1u1, version 26.0.1.25688693 (Figures A1a, A1b).	Broadcom Support Portal, file VMware-Workstation-Full-26H1u1-25688693.exe (267.61 MB, release date Sep 03, 2026 as displayed).
Check Point SmartConsole	R82, build 82.0.9800.1071 in the About window (Figure A1c). Login and fingerprint screens show build 82.0.9800.1027 (Figures A6d, A6e).	Supplied by the instructor.
A-SMS virtual machine	Gaia R82, build 779 (Figure G1)	Instructor-supplied package, A-SMS.vmx.

The About dialog for Workstation Pro 26.0.1 also shows the A-SMS.vmx location, which ties this installation to the lab. Two SmartConsole build numbers appear, both R82. The About window says updates are enabled, and the client updated itself automatically between the login screens (1027) and the About capture (1071). This is recorded as confirmed by me, and it did not affect any step.



Figure A1a. *VMware Workstation Pro 26H1u1, version 26.0.1.25688693, with the A-SMS.vmx location shown, proving this is the version used for the lab.*

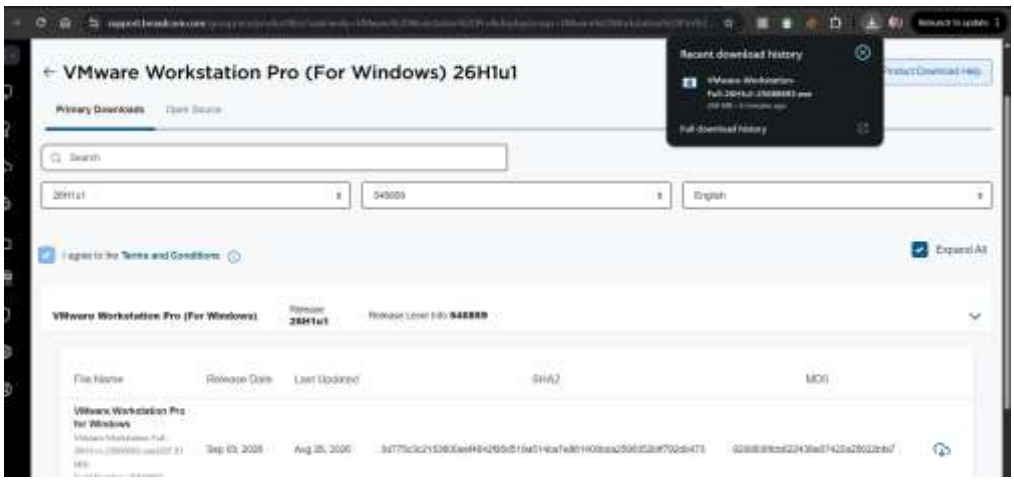


Figure A1b. *Broadcom Support Portal download page for VMware Workstation Pro 26H1u1 (Windows) and the downloaded installer file.*

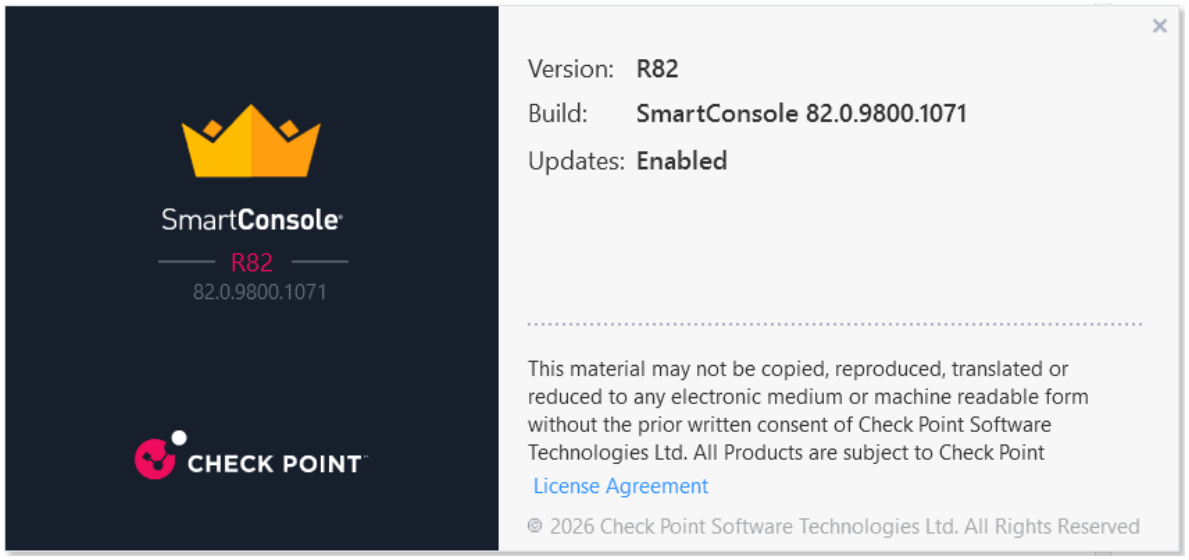


Figure A1c. *SmartConsole About window: Version R82, build 82.0.9800.1071, Updates enabled.*

A2 A-SMS virtual machine and VMXNET3 adapter

A-SMS was opened from A-SMS.vmx and kept powered off while it was edited. The hardware summary (Figure A2a) shows 8 GB memory, 8 processors and a 500 GB disk. The course guide recommends 12 GB RAM with 8 GB as the minimum, so the 8 GB allocation meets the minimum but not the recommendation. The host reports 15,708 MB of memory (Figure A1a), which is why a larger allocation was not used. Processor count and disk size match the guide.

Before the change the adapter showed NAT (Figure A2a). It was set to Custom with VMnet8 selected and connect at power on ticked (Figure A2b). The Connected box is greyed because the VM was off. The VMX excerpt (Figure A2c) shows ethernet0.connectionType = "custom" and ethernet0.virtualDev = "vmxnet3", and only ethernet0 entries appear, so a single adapter is configured. VMXNET3 is the paravirtualised adapter type that this Gaia image needs in order to detect eth0.

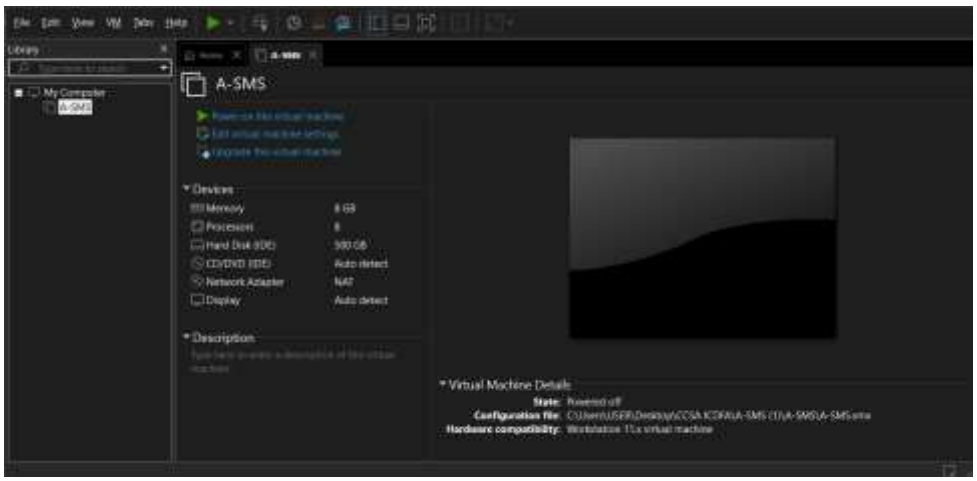


Figure A2a. *A-SMS hardware summary before the adapter change: 8 GB memory, 8 processors, 500 GB disk, powered off, network adapter shown as NAT.*

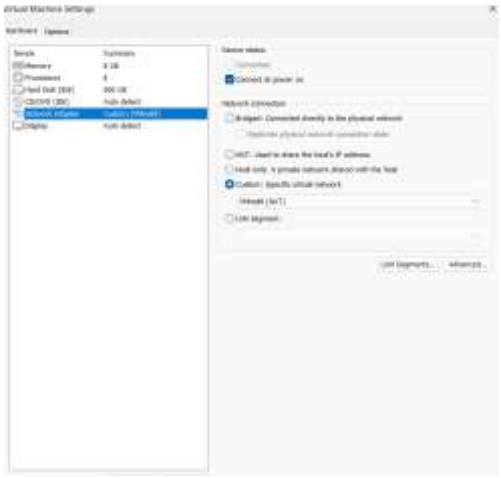


Figure A2b. *Virtual Machine Settings: network adapter set to Custom (VMnet8) with Connect at power on ticked.*

```
.encoding = "UTF-8"
config.version = "8"
virtualHW.version = "11"
memsize = "8192"
MemAllowAutoScaleDown = "FALSE"
displayName = "A-SMS"
guestOS = "other"
numvcpus = "8"
cpuid.coresPerSocket = "8"
firmware = "bios"
ethernet0.present = "TRUE"
ethernet0.addressType = "generated"
ethernet0.connectionType = "custom"
ethernet0.virtualDev = "vmxnet3"
ide0:0.present = "TRUE"
ide0:0.fileName = "A-SMS.vmdk"
ide1:0.present = "TRUE"
ide1:0.autodetect = "TRUE"
ide1:0.filename = "auto detect"
ide1:0.deviceType = "cdrom-raw"
mks.enable3d = "TRUE"
vmci0.present = "TRUE"
pciBridge0.present = "TRUE"
pciBridge4.present = "TRUE"
pciBridge4.virtualDev = "pcieRootPort"
pciBridge4.functions = "8"
```

Figure A2c. *A-SMS.vmx opened in a text editor with the line ethernet0.virtualDev = "vmxnet3" highlighted.*

A3 VMnet8 NAT plan

Setting	Required	Observed	Figure
Network type	NAT, host adapter connected, local DHCP on	NAT; host connection Connected; DHCP Enabled; host adapter VMware Network Adapter VMnet8	A3a
Subnet / mask	10.1.1.0 / 255.255.255.0	10.1.1.0 / 255.255.255.0	A3a, A3b
NAT gateway	10.1.1.250	10.1.1.250	A3b
DHCP pool	10.1.1.210 to 10.1.1.240	10.1.1.210 to 10.1.1.240 (broadcast 10.1.1.255; default lease 30 minutes, maximum 2 hours)	A3c

The Virtual Network Editor also lists VMnet0 (bridged) and VMnet1 (host-only, 192.168.80.0). These are outside the lab plan and were not changed. The two static addresses used later (10.1.1.101 and 10.1.1.201) lie outside the DHCP pool.

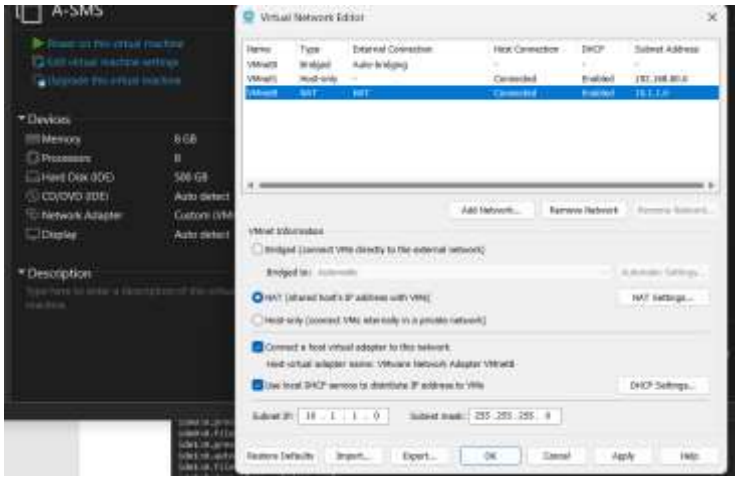


Figure A3a. *Virtual Network Editor: VMnet8, type NAT, host virtual adapter connected, local DHCP enabled, subnet 10.1.1.0 / 255.255.255.0.*

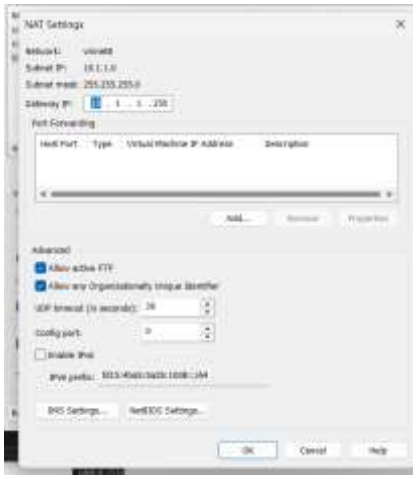


Figure A3b. *VMnet8 NAT Settings: gateway IP 10.1.1.250.*

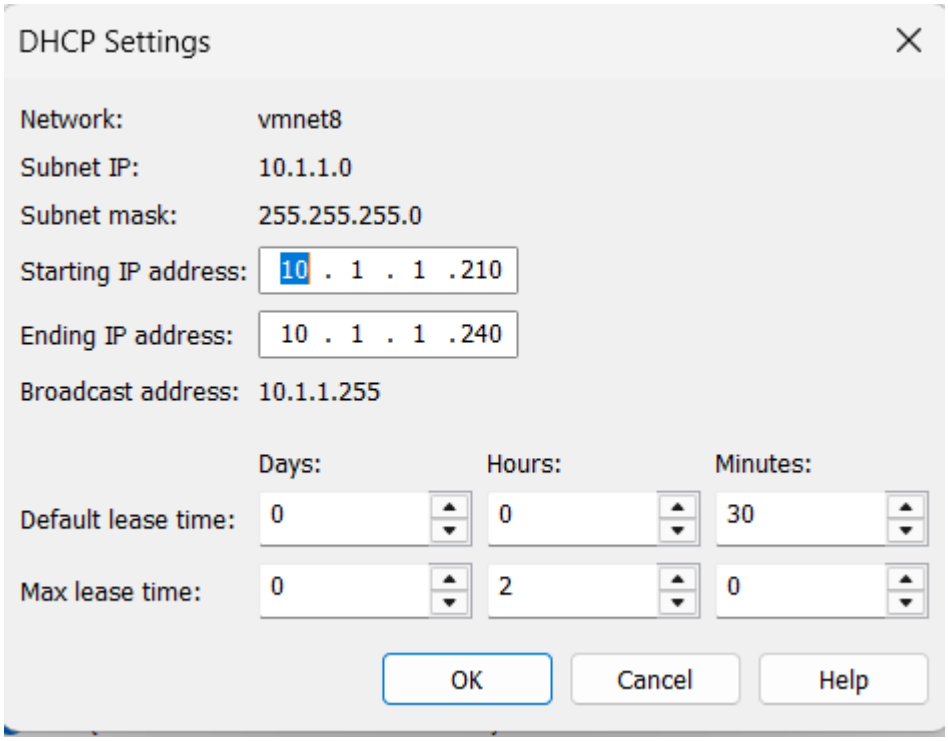


Figure A3c. *VMnet8 DHCP Settings: pool 10.1.1.210 to 10.1.1.240.*

A4 Windows VMnet8 address

The VMnet8 adapter was addressed from an administrator Command Prompt using netsh interface ipv4 set address name="VMware Network Adapter VMnet8" source=static address=10.1.1.201 mask=255.255.255.0 gateway=none, followed by ipconfig. Figure A4 shows the resulting state: 10.1.1.201, mask 255.255.255.0 and no default gateway. The screenshot captures the ipconfig result rather than the netsh command itself. The other adapters in the listing (Bluetooth, disconnected, and vEthernet Default Switch, 172.18.48.1) are not part of the lab, and the physical Wi-Fi and Ethernet settings were not changed.

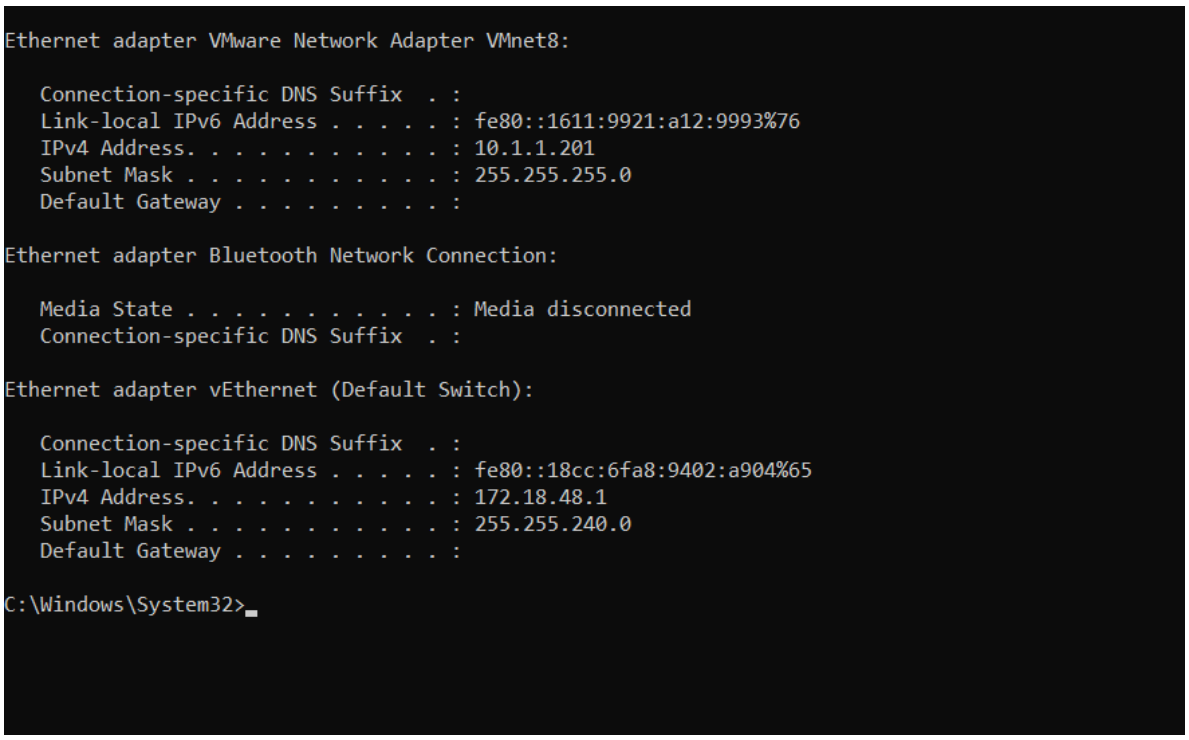


Figure A4. *ipconfig output: VMware Network Adapter VMnet8 at 10.1.1.201, mask 255.255.255.0, no default gateway.*

A5 A-SMS eth0 configuration

After power-on, the Gaia Clish session on the VMware console was used to run show interface eth0 (Figure A5). The output shows **state on**, **link-state link up**, **ipv4-address 10.1.1.101/24**, MAC 00:0c:29:69:24:ce, speed 10G, full duplex and MTU 1500. The address and state were already correct when first inspected, and the evidence shows no set interface or save config command. No correction was needed.



Figure A5. *Gaia Clish: show interface eth0 reports state on, link-state link up and ipv4-address 10.1.1.101/24.*

A6 Connectivity, Gaia Portal and SmartConsole access

Ping. ping 10.1.1.101 from Windows returned 4 replies out of 4 sent, 0% loss, 1 to 3 ms, TTL 64 (Figure A6a). A TTL of 64 is typical of a Linux-based host such as Gaia.

Gaia Portal. https://10.1.1.101 loaded the Gaia Portal R82 login page (Figure A6c). The browser showed Not secure because the portal uses a self-signed certificate. The address is the one configured on my own A-SMS eth0 (Figure A5), so the warning was accepted.

Fingerprint check. SmartConsole was started with the account cpadmin and server 10.1.1.101 (Figure A6d). On the first connection it displayed an ICA fingerprint (Figure A6e). Without using that prompt as the source of truth, I retrieved the fingerprint locally on the A-SMS VMware console, a channel I trust, with expert, cp_conf finger get and exit (Figure A6f). Both values read HER PAR BERT NOD BUST DOWN FOOT SANG EGG SAY BASE FOLD, word for word, so the connection was trusted and I proceeded. The Gaia account (admin) and the SmartConsole account (cpadmin) serve different interfaces.

Authenticated session. Figure A6g shows SmartConsole connected to 10.1.1.101 as cpadmin. A-SMS has a green status, IP 10.1.1.101 and version R82. Its licence and alert status appear in Figure S1a. **Part A checkpoint:** A-SMS was reachable through the Gaia Portal and through an authenticated SmartConsole session.

```
C:\Windows\System32>ping 10.1.1.101

Pinging 10.1.1.101 with 32 bytes of data:
Reply from 10.1.1.101: bytes=32 time=1ms TTL=64
Reply from 10.1.1.101: bytes=32 time=3ms TTL=64
Reply from 10.1.1.101: bytes=32 time=2ms TTL=64
Reply from 10.1.1.101: bytes=32 time=1ms TTL=64

Ping statistics for 10.1.1.101:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms
```

Figure A6a. *Windows ping to 10.1.1.101: 4 sent, 4 received, 0% loss.*

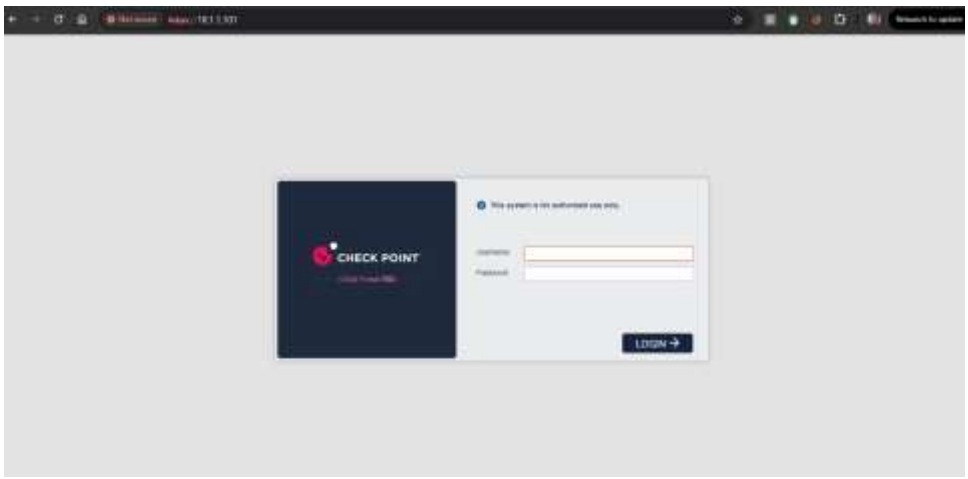


Figure A6c. *Gaia Portal R82 login page loaded at https://10.1.1.101 (browser marks it Not secure).*

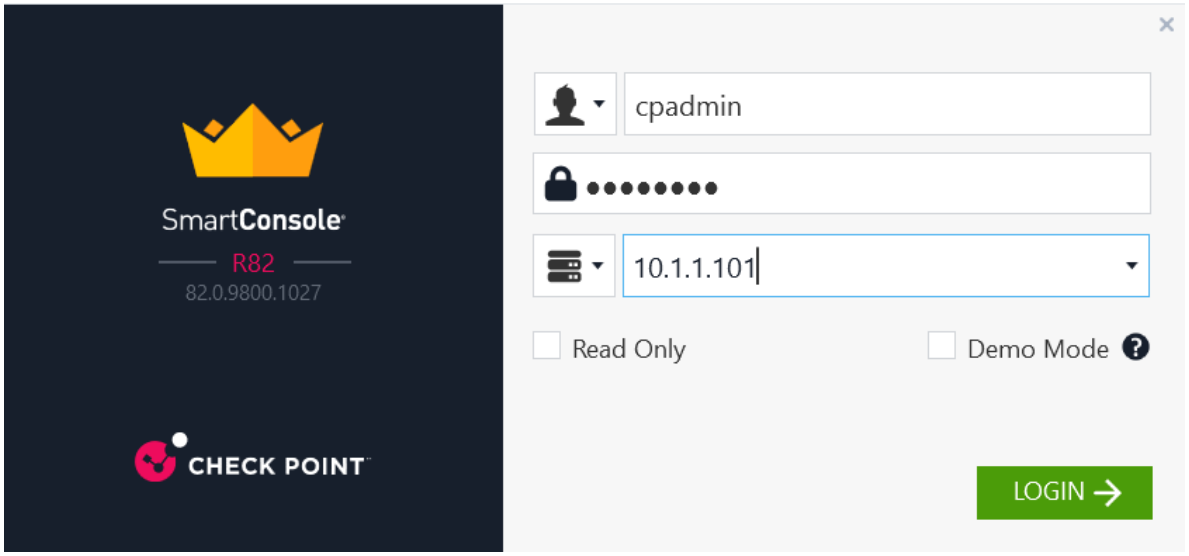


Figure A6d. *SmartConsole login: account cpadmin, server 10.1.1.101 (password masked).*

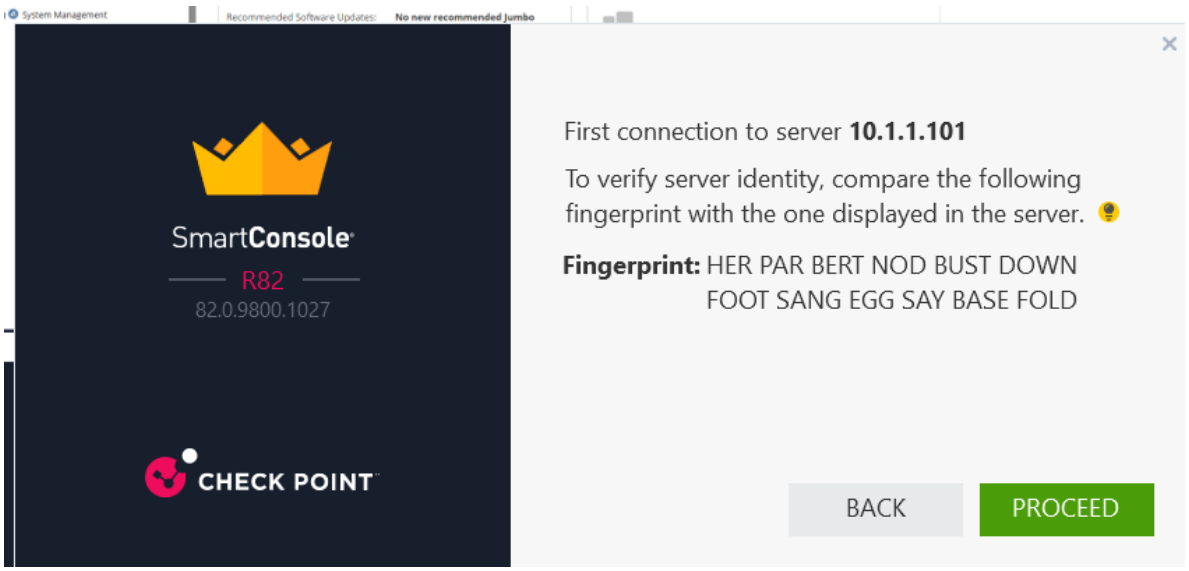


Figure A6e. *SmartConsole first-connection prompt showing the server fingerprint HER PAR BERT NOD BUST DOWN FOOT SANG EGG SAY BASE FOLD.*

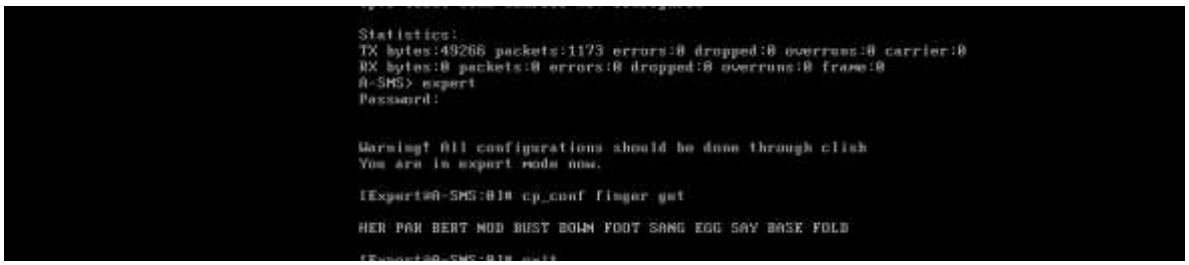


Figure A6f. *A-SMS VMware console: expert, then cp_conf finger get returns the identical fingerprint; exit follows.*

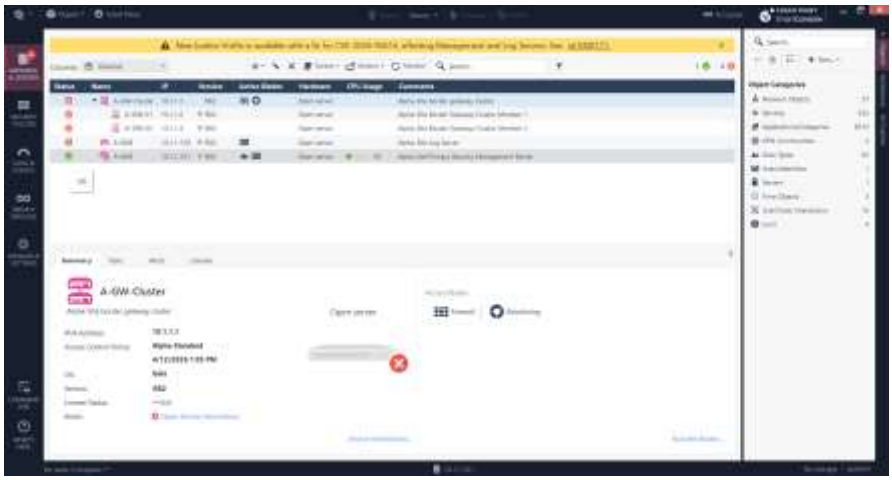


Figure A6g. *Authenticated SmartConsole session (cpadmin on 10.1.1.101) in Gateways & Servers; A-SMS has a green status.*

Part B: Theory Answers

B1 Explain the architecture

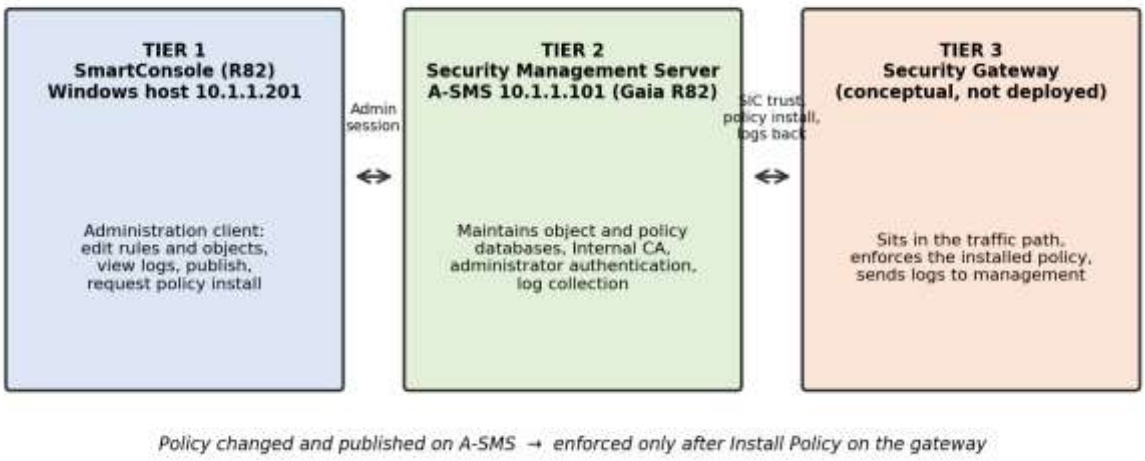


Figure B1. *Three-tier Check Point architecture used in this lab.*

SmartConsole is the first tier. It runs on my Windows host (10.1.1.201 on VMnet8) and is only an administration tool: it does not store the policy or configuration, but sends them to the Security Management Server for storage (slides 4, 7). The Security Management Server, A-SMS at 10.1.1.101, is the second tier. It keeps the object database and policies, hosts the log files and runs the Internal Certificate Authority (slides 24 to 29). The Security Gateway is the third tier and the first line of defence: it sits in the traffic path and inspects inbound and outbound traffic (slide 40). In this lab the gateway is conceptual; A-GW-Cluster shows a red status in Figure A6g and was not deployed.

Changing a policy means editing rules in a session on the management server, which does not affect the live environment (slide 19). Enforcing it means installing it on a gateway. Until then, the gateway keeps enforcing the last installed policy.

B2 Explain management services and deployment

Three Security Management Server functions are shown on A-SMS in Figure S1a. The database securely stores managed data, including policies and objects, and serves it on request (slide 25); this is the Network Policy Management feature (slide 38). The log server hosts log files collected from gateways (slide 26) and appears as Logging & Status. The Internal Certificate Authority creates certificates for users and administrators (slide 29), which is why SmartConsole asked me to verify the ICA fingerprint on first connection (Figure A6e).

An on-premises deployment, as I built here on an open server running Gaia (slide 35), means I provision and maintain the server, in return for control of data location and upgrade timing. Smart-1 Cloud is management delivered as a service; its typical use case is a company wanting greater efficiency and less complexity in its management platform (slide 36). For an organisation that wants less on-premises hardware maintenance I

recommend Smart-1 Cloud. The tradeoff is reduced control: it relies on the provider's schedule for platform changes and on connectivity to the service.

B3 Choose the correct administration interface

Task	Tool	Reason
Inspect eth0 and its subnet mask	Gaia Portal (Network Interfaces) or Clish show interface eth0	Interfaces belong to Gaia, not the policy database (slide 60). The Portal showed 255.255.255.0 and Clish /24 (Figures G3, A5).
Review an Access Control rule	SmartConsole, Security Policies	SmartConsole is where policies are created and managed (slide 12); rules show Source, Destination, Services and Action (Figure S2a).
Inspect operating-system resource use	Expert-mode tools such as top; SmartConsole device summary for a quick view	Expert is the Linux shell (slides 56, 57). The Gaia Overview here shows no CPU or memory, but SmartConsole monitoring reported both (slide 28, Figure S1a).
View security logs	SmartConsole, Logs & Events	This view shows events and logs (slide 12; Figure S4a).

Gaia Clish is reached on the console or over SSH, uses commands in the form operation feature parameter, and shows the prompt A-SMS> (slides 55, 58). Expert mode is entered with expert and its own password and left with exit (slides 56, 57); its prompt is [Expert@A-SMS:0]# (Figure G7a), and it allows Linux commands such as cp_conf, with greater risk. SmartConsole's Command Line panel runs API commands and scripts from the Security Management Server (slide 13). It is neither Gaia shell, which is why ip route and ping -c were rejected there (Figure C8).

B4 Explain security features and policy handling

Firewall and Application Control are Access Control features (slide 47). Firewall inspects traffic and drops or blocks what policy does not allow: a company can allow only web and DNS from its office subnet and block unsolicited inbound connections to its server network. Application Control, with URL Filtering, detects and controls applications and sites by category: a school can allow general browsing but block streaming or gambling categories.

Publishing makes session changes visible to other administrators, and unpublished changes are not installed during a policy installation (slide 19). Installing sends the published policy to the chosen gateways, which then enforce it (slides 17, 18). They are separate because a session is a working area where I can change things without affecting the live environment: I can review the changes, discard them or publish them, and install

when ready. Neither was performed in this lab: Publish was greyed out and the status bar read No changes (Figures A6g, C13).

B5 Interpret your own lab evidence

VMnet8 is a virtual NAT network that exists only inside my computer. Windows (10.1.1.201) and A-SMS (10.1.1.101) communicate across it, and VMware's NAT service forwards outside traffic through whichever physical connection is active. The 10.1.1.0/24 subnet is defined in the Virtual Network Editor rather than learned from a router, so changing Wi-Fi does not renumber it (Figure A3a). VMware DHCP only offers leases from 10.1.1.210 to 10.1.1.240, and only to clients that ask (Figure A3c). A-SMS has a static address set in Gaia and does not request a lease, and both static addresses lie outside the pool, so DHCP can neither assign nor replace them.

A successful ping (Figure A6a) shows only that Windows can exchange ICMP echo packets with 10.1.1.101; because ICMP can be filtered, a failed ping would not prove that services are down. A SmartConsole login proves more: the management services are running and reachable, the server identity was checked by fingerprint, and the cpadmin account was authenticated (Figures A6e to A6g).

An unavailable gateway object or an empty or partial log view is a status, not a diagnosis. Here A-GW-Cluster and the log server object show red (Figure A6g), consistent with only A-SMS being deployed, and the log search warned that log server A-SME was disconnected (Figures S4a, S4c). Before diagnosing a fault I would need the exact alert text, log source, time range, device state, and whether traffic could have produced logs.

Part C: Observation Record

Each task lists the menu path or command, the observed facts, a separate interpretation and the figure. Observations come from the screenshots. Where a screenshot does not show a value, that is stated and nothing is inferred. All inspection was read-only.

Gaia Portal observations (C1 to C5)

C1 System overview

Menu path / command	Gaia Portal (https://10.1.1.101) > Overview
Observation	Host name A-SMS (header); product Check Point Security Management; version R82; kernel 4.18.0-372.9.1cpx86_64; edition 64-bit; build number 779; system uptime 17 minutes; platform Open Server; no new recommended Jumbo Hotfix detected. Network Configuration lists eth0 10.1.1.101 (Up) and lo 127.0.0.1 (Up). The Gaia Portal Overview in this build has no CPU or memory panel (confirmed by scrolling the page), so those two values cannot be reported from Gaia. SmartConsole reports CPU 3% and memory 47% for A-SMS (Figure S1a).
Interpretation	Resource readings change because processes start, stop and cache data, so each value is a snapshot of one moment. Uptime reads 17, 32 and 48 minutes in Figures G1, G2a and G6a, which fits a single boot observed at later times.
Figure	G1

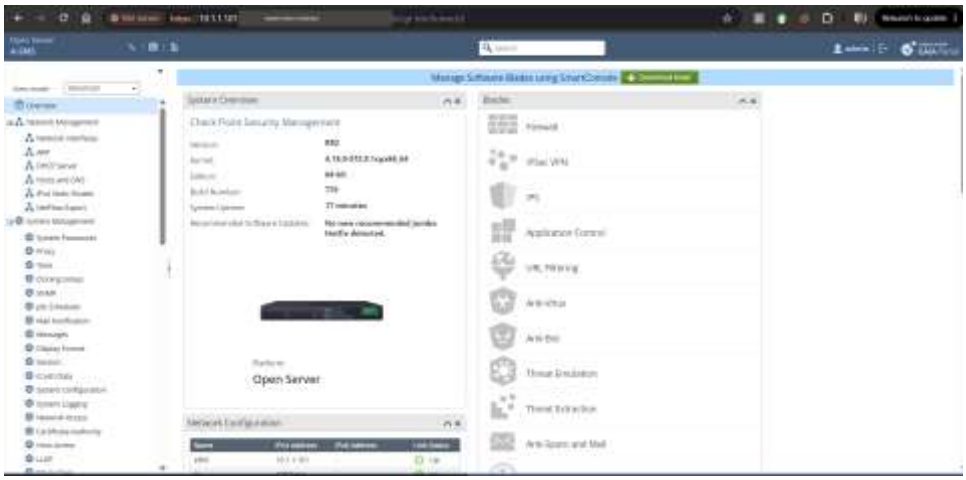


Figure G1. *Gaia Portal Overview: A-SMS, R82, kernel 4.18.0-372.9.1cpx86_64, build 779, uptime 17 minutes (session token in URL redacted).*

C2 Navigation modes

Menu path / command	Gaia Portal > View mode drop-down (Basic / Advanced)
Observation	Basic view lists Network Management (Network Interfaces, ARP, Hosts and DNS, IPv4 Static Routes), System Management, User Management, High Availability, Maintenance and Software Updates (Figure G2a). Advanced view (shown in Figures G1 and G2b) adds further pages, including Advanced Routing (DHCP Relay, BGP, IGMP, OSPF, RIP and others), DHCP Server, SNMP and Job Scheduler. Three navigation areas: Network Management (interfaces, hostname, DNS, routes); System Management (time, passwords, messages, system configuration); Maintenance (licence status, snapshots, backup, shut down). Advanced-only example: Advanced Routing, which does not appear in Basic view.
Interpretation	Basic view keeps everyday settings visible, while Advanced exposes specialist features such as dynamic routing protocols. Both views read from the same system (slide 62).
Figure	G2a, G2b



Figure G2a. *Gaia Portal in Basic view mode (uptime 32 minutes).*

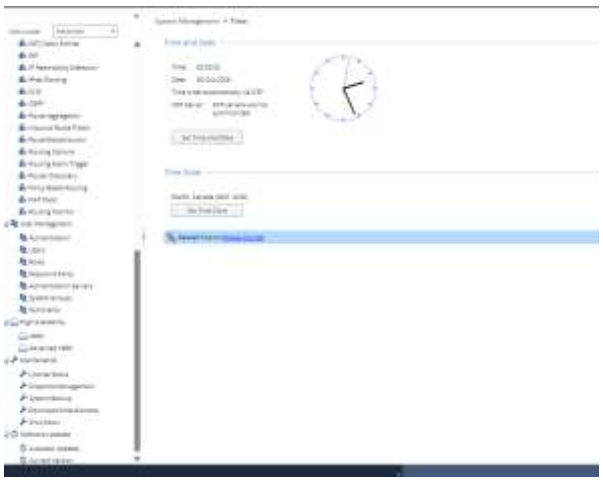


Figure G2b. *Gaia Portal in Advanced view mode, menu scrolled to Advanced Routing, User Management, High Availability and Maintenance.*

C3 Management interface

Menu path / command	Gaia Portal > Network Management > Network Interfaces
Observation	eth0: type Ethernet, IPv4 10.1.1.101, subnet mask 255.255.255.0, link status Up. lo: type Loopback, 127.0.0.1, mask 255.0.0.0, link Up. The table has no MAC or administrative-state column, and no Edit dialog appears in the capture. MAC 00:0c:29:69:24:ce and state on come from Clish (Figures A5, G6c).
Interpretation	The Portal, the Part A Clish output and the Part A plan agree: 10.1.1.101/24 is the same as 255.255.255.0. The Portal reports the mask in dotted form, Clish as a prefix length.
Figure	G3 (cross-reference A5)



Figure G3. *Network Management > Network Interfaces: eth0 Ethernet 10.1.1.101, 255.255.255.0, link Up; lo Loopback 127.0.0.1.*

C4 Routes and DNS

Menu path / command	Gaia Portal > Network Management > IPv4 Static Routes; Network Management > Hosts and DNS

Observation	Static route: Destination Default, next hop type Normal, rank 60, gateway 10.1.1.1 (priority None), monitored protocols None, ping No. Hosts and DNS: host name A-SMS, domain alpha.cp, DNS suffix alpha.cp, primary DNS 168.63.129.16, secondary 192.168.11.101, tertiary 1.1.1.1. The default gateway (10.1.1.1) differs from the VMware NAT gateway (10.1.1.250, Figure A3b). The values were recorded and not changed. No test of internet access from A-SMS was performed or recorded, so internet reachability is unverified.
Interpretation	A default gateway is the router that receives traffic for destinations outside the local subnet. A DNS resolver translates names into IP addresses. They are separate services, and either can fail on its own. Local SmartConsole access over 10.1.1.0/24 does not need either, so it does not show that the VM can reach the internet. Hypothesis, not tested: with the default route at 10.1.1.1 rather than 10.1.1.250, outbound internet traffic would not be expected to work unless something answers at 10.1.1.1.
Figure	G4a, G4b

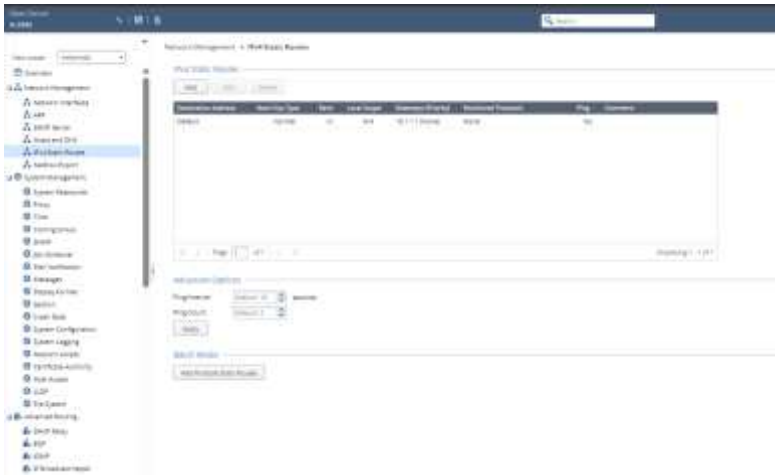


Figure G4a. *Network Management > IPv4 Static Routes: one Default route via 10.1.1.1, rank 60.*

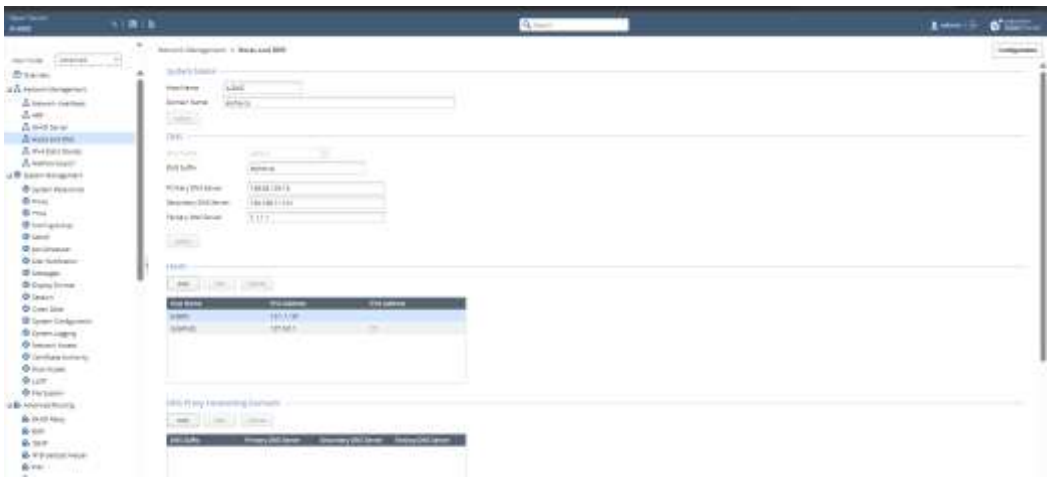


Figure G4b. *Network Management > Hosts and DNS: host A-SMS, domain alpha.cp, DNS servers 168.63.129.16, 192.168.11.101 and 1.1.1.1.*

C5 Maintenance and administration functions

Menu path / command	User Management > Authentication / Users / Roles / GUI Clients; System Management > Time; Maintenance > Snapshot Management; Maintenance > System Backup
Observation	User Management: Authentication, Users, Roles and GUI Clients in Basic view (Figure G2a); Advanced view adds Password Policy, Authentication Servers and System Groups (Figure G2b). Time: 02:24:56, 30-Oct-2026, set automatically via NTP, NTP servers not synchronized; time zone Pacific, Canada (GMT -8:00) (Figure G5a). Snapshot Management: My Snapshot list empty, scheduled snapshot disabled, creating an image needs 16.644G and 166.33G is available (Figure G5b). System Backup: one local backup, backup_--_A-SMS.alpha.cp_2025_07_09_04_53_56.tgz, 185.21MB, R82 (take 777), source Manual, role Security Management (Primary), location /var/log/CPbackup/backups; scheduled backups empty (Figure G5c). The backup is dated before this lab. No backup, snapshot, update or restore was run.
Interpretation	User Management controls Gaia operating-system accounts and roles. Time and NTP keep clocks consistent, which matters for logs and certificates; the unsynchronised state was recorded and not corrected. A configuration backup is an archive of Check Point and Gaia settings and data that is restored onto a compatible installation. A system snapshot is an image of the whole system (operating system and installed software) that can be reverted to, and needs far more disk space. This is why the snapshot page quotes 16.644G while the backup is 185.21MB.
Figure	G5a, G5b, G5c (menu paths also in G2a, G2b)

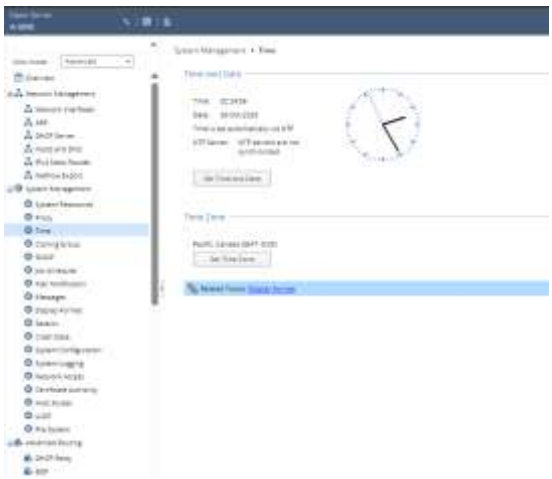


Figure G5a. *System Management > Time: NTP configured but not synchronised; time zone Pacific, Canada (GMT -8:00).*

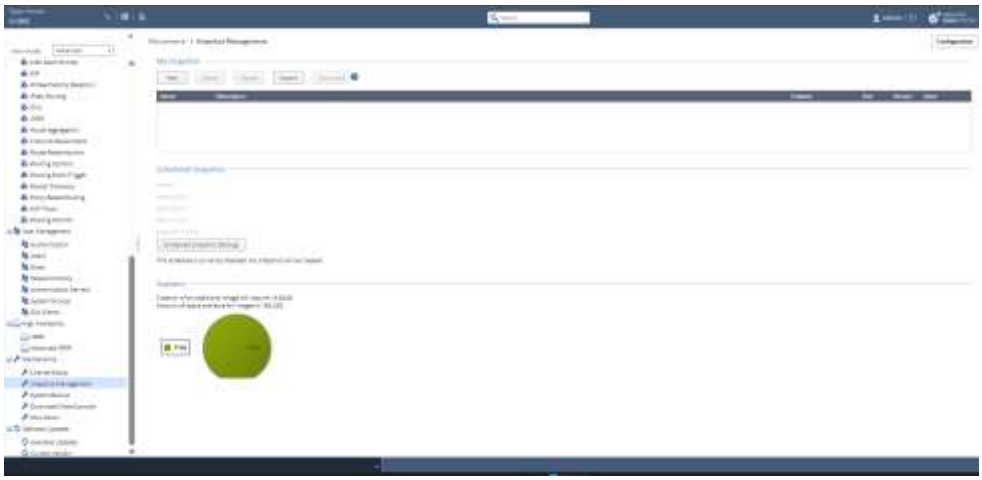


Figure G5b. *Maintenance > Snapshot Management: no snapshots listed, scheduled snapshot disabled.*

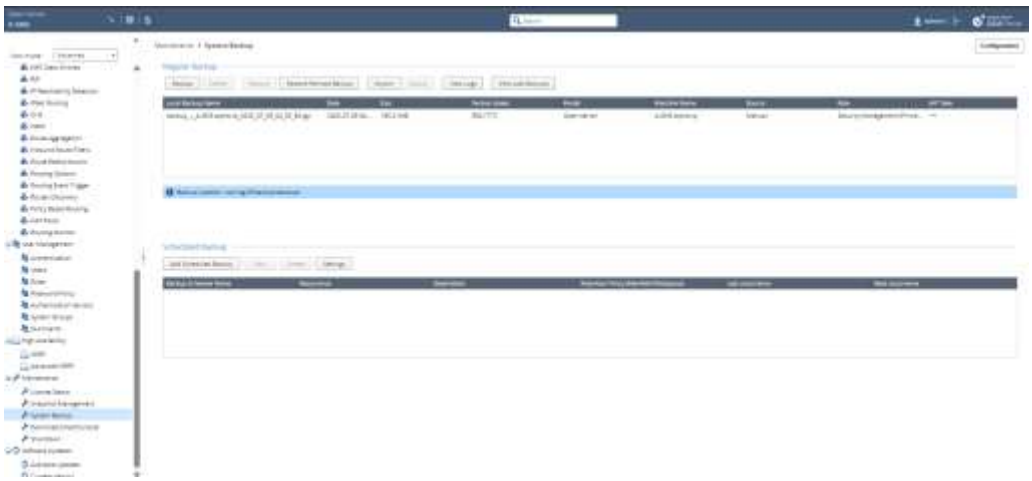


Figure G5c. *Maintenance > System Backup: one existing local backup listed; scheduled backups empty.*

Gaia Clish and Expert mode (C6 to C8)

C6 System information in Clish

Menu path / command	VMware console > Gaia Clish: show version all, show uptime, show interfaces all, show interface eth0
Observation	show version all: Product version Check Point Gaia R82, OS build 779, kernel 4.18.0-372.9.1cpx86_64, edition 64-bit. show uptime: 48 minutes. show interfaces all: eth0 with state on, MAC 00:0c:29:69:24:ce, link up, MTU 1500, speed 10G, full duplex, 10.1.1.101/24, TX 96,171,188 bytes and RX 7,009,390 bytes with no errors or drops (Figure G6b); then lo with state on, no MAC, type loopback, link-state not available, MTU 65536, 127.0.0.1/8 (Figure G6c). show interface eth0: the same eth0 values with TX 35,373,852 bytes and RX 2,866,736 bytes, captured earlier (Figure G6d).
Interpretation	show version all identifies the product release and build; show uptime shows time since boot; show interfaces all lists every interface with its attributes; show interface eth0 gives one interface in detail. The counters differ between captures because they were taken at different times. Release R82 and build 779 match the Portal (Figure G1), and eth0 10.1.1.101 matches Figure G3. Clish prints plain key and value lines, with the address as a /24 prefix and extra detail (MAC, speed, counters). The Portal presents a table with the mask in dotted form and a graphical link indicator.
Figure	G6a, G6b, G6c, G6d (A5 also shows eth0)



Figure G6a. *Clish: show version all and show uptime.*

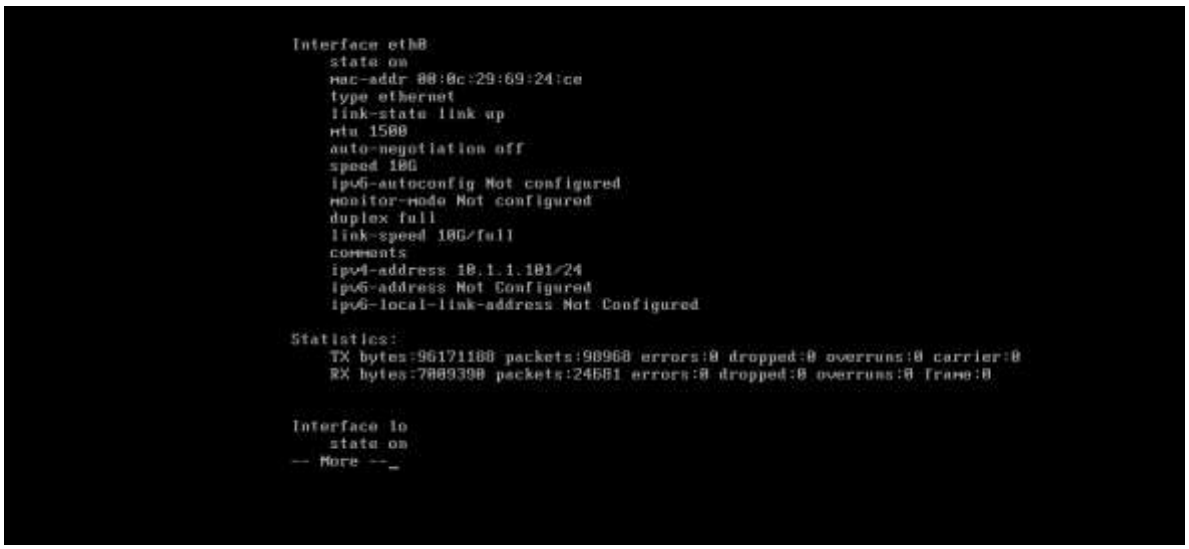


Figure G6b. *Clish: show interfaces all, eth0 section (output paused at -- More --).*

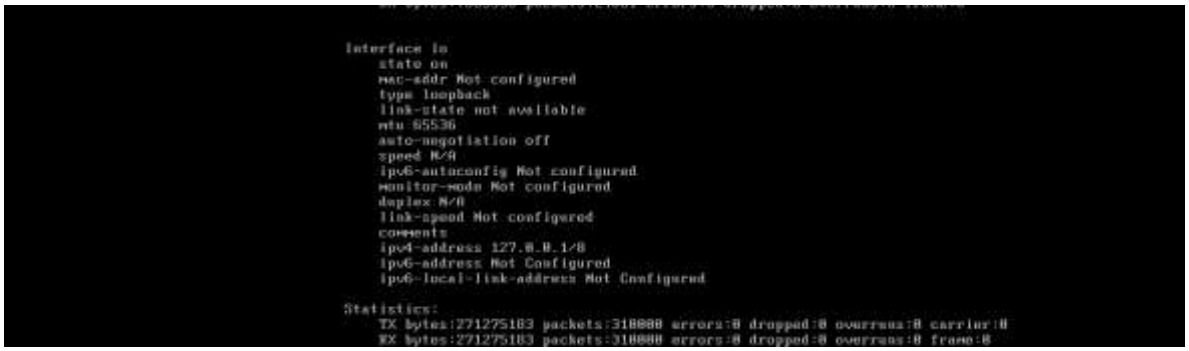


Figure G6c. *Clish: show interfaces all, continued with the lo section.*

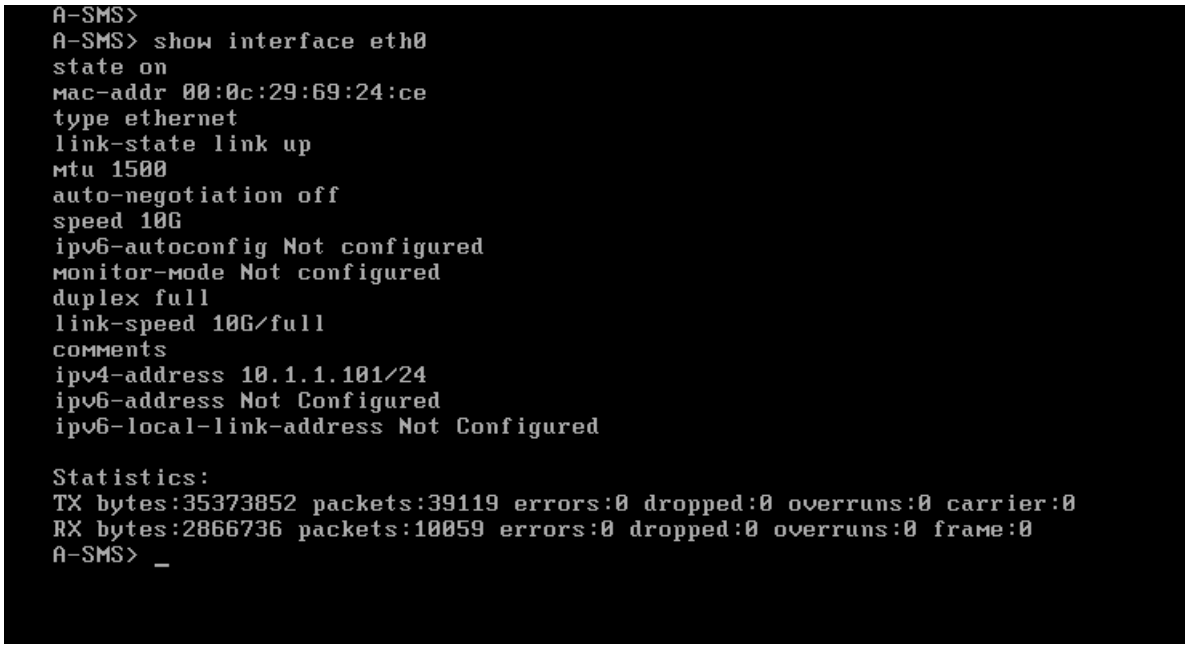


Figure G6d. *Clish: show interface eth0.*

C7 Expert shell

Menu path / command	VMware console: expert, whoami, pwd, exit
Observation	After the password prompt, a warning states that all configurations should be done through clish, and the prompt changes from A-SMS> to [Expert@A-SMS:0]#. whoami returned admin and pwd returned /home/admin (Figure G7a). In a later session, exit returned the prompt to A-SMS> (Figure G7b). That login also displayed the message "CLINFR0771 Config lock is owned by admin. Use the command 'lock database override' to acquire the lock." The override was not used.
Interpretation	Clish is a restricted command interpreter that offers only Gaia's own show and set commands. Expert mode is the underlying Linux Bash shell, so ordinary Linux commands run there, with greater risk of unintended changes. The warning reflects that. The config-lock message means another admin session holds Gaia's configuration lock, most likely my Gaia Portal login (hypothesis, not tested). It did not affect read-only inspection.
Figure	G7a, G7b (cross-reference A6f)

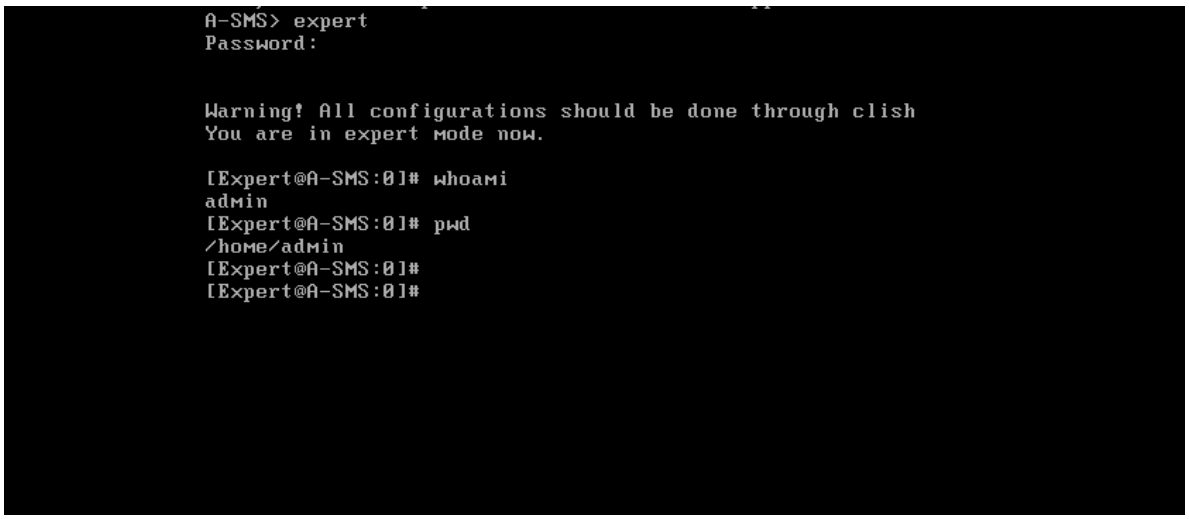


Figure G7a. *Expert mode: prompt [Expert@A-SMS:0]#, whoami returns admin, pwd returns /home/admin.*

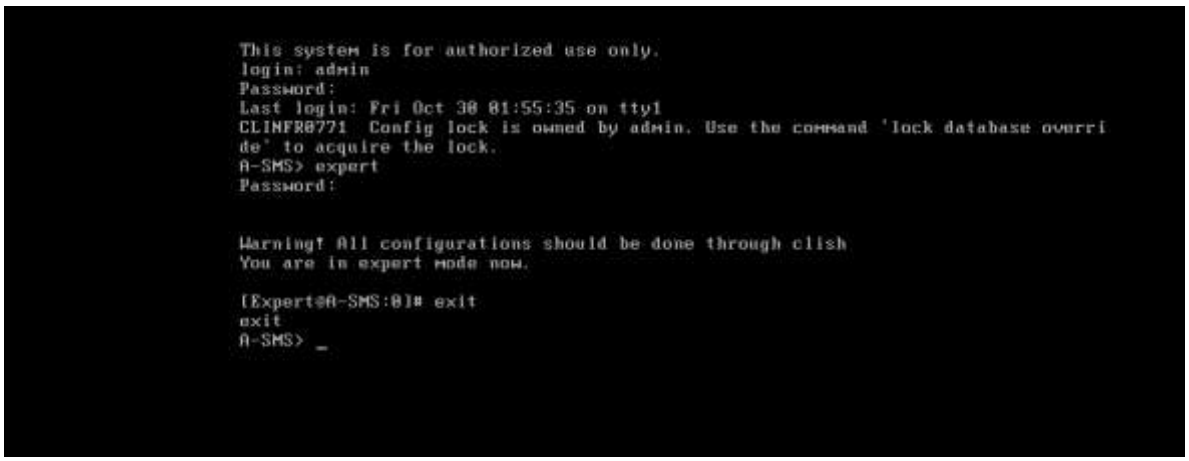


Figure G7b. *Clish login, config-lock message, expert, then exit returning to the A-SMS> prompt.*

C8 Command environment

Menu path / command	SmartConsole > Command Line (bottom of the left rail)
Observation	The Command Line window opened as "Command Line (10.1.1.101)". Entering ip route returned an error that the parameters of the ip command must be given in key and value pairs. Entering ping -c returned that the flag -c is not supported within SmartConsole's CLI window. The brief says viewing the panel is sufficient; these two inputs were only entered to observe the rejection, and no management command or script was run.
Interpretation	The panel does not give a Linux shell. It runs API commands and scripts from the Security Management Server (slide 13), so Linux-style ip route and ping -c are rejected. Linux diagnostics belong in an Expert session on the server's VMware console, where a real Bash shell is available.
Figure	C8

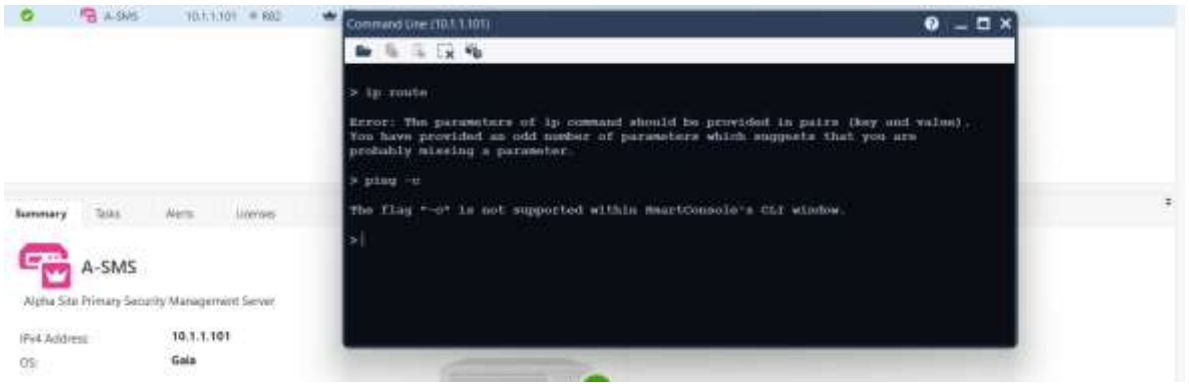


Figure C8. *SmartConsole Command Line window: ip route and ping -c are rejected.*

SmartConsole observations (C9 to C13)

C9 Gateways and Servers

Menu path / command	SmartConsole > Gateways & Servers > A-SMS > Summary, Alerts, Licenses
Observation	A-SMS: IPv4 10.1.1.101, OS Gaia, version R82, "No Jumbo Hotfix information is available", status green, licence status OK, alerts OK, CPU 3%, memory 47%. Management blades: Network Policy Management and Logging & Status. Alerts tab: "No Errors". Licenses tab: four entries for 10.1.1.101; two show expiration 30/09/2026 and two show Never (CK values redacted). Banner: a new Jumbo Hotfix is advertised with a fix for CVE-2026-93616 affecting Management and Log Servers (sk1000171). Other devices: A-GW-Cluster (10.1.1.1), A-GW-01 (10.1.1.2), A-GW-02 (10.1.1.3) and A-SME, a log server (10.1.1.130), all R82 with a red status. The top-right counters read one green and four red. A-GW-Cluster lists Alerts as an error with an Open Device Information link, and its Access Control Policy is shown as Alpha-Standard.
Interpretation	A-SMS is healthy as far as SmartConsole reports. The red devices fit the lab design, in which only A-SMS runs, and I did not investigate or change them. Two licence entries carry an expiry date earlier than the completion date (30/10/2026) while the summary still says OK. I cannot tell from the screenshots why; a possible reason, not tested, is that the remaining non-expiring entries cover the server. The Gaia overview reports no new recommended Jumbo Hotfix, while the SmartConsole banner advertises one. Both statements are recorded as displayed.
Figure	S1a, S1b, S1c (device list in A6g)

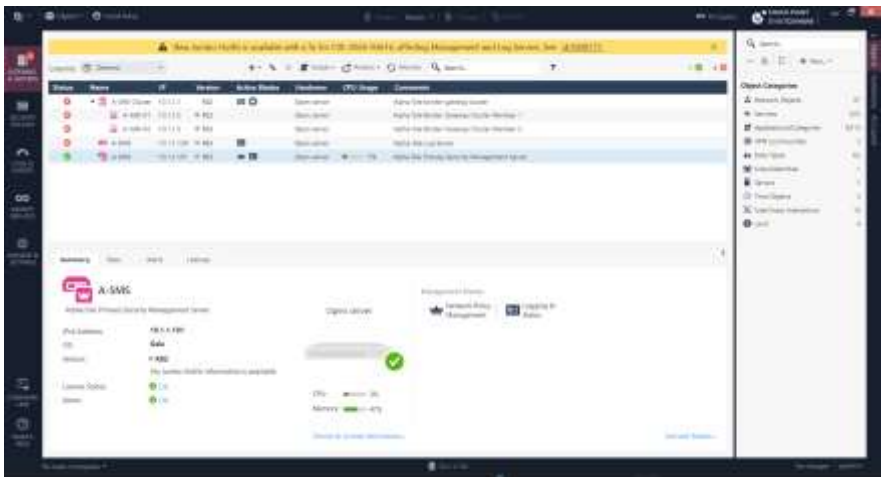


Figure S1a. *SmartConsole Gateways & Servers, A-SMS Summary: 10.1.1.101, Gaia, R82, licence OK, alerts OK, CPU 3%, memory 47%.*

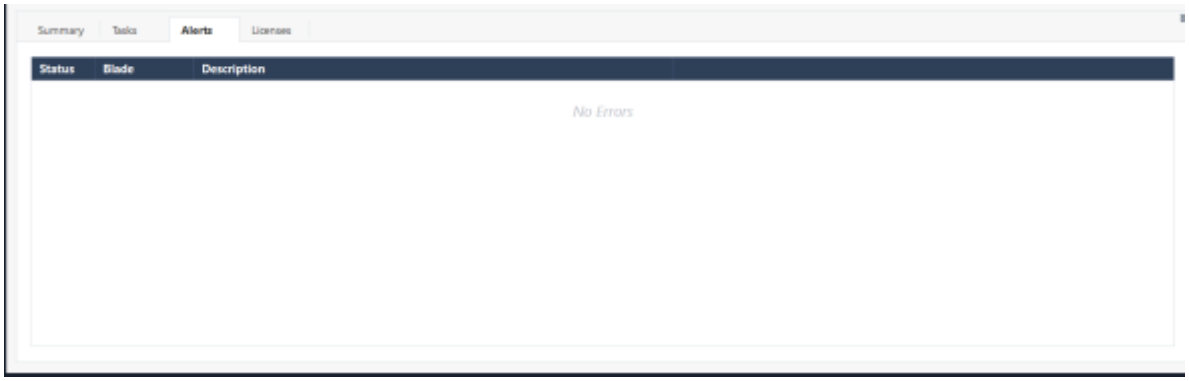


Figure S1b. *A-SMS Alerts tab: No Errors.*

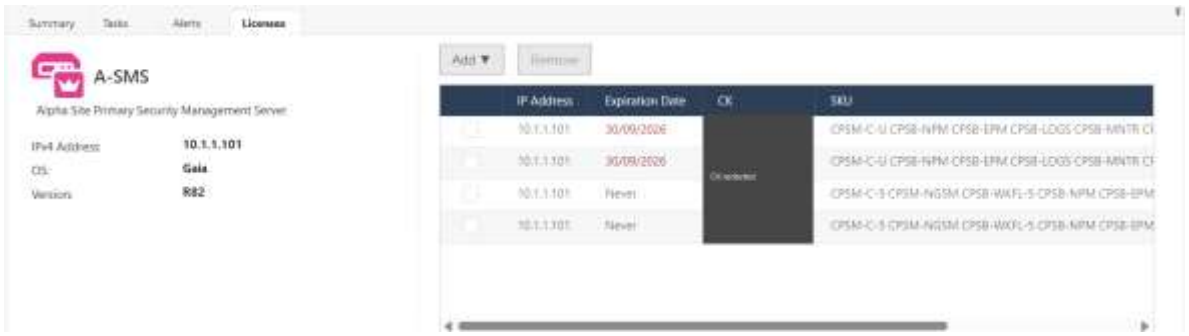


Figure S1c. *A-SMS Licenses tab: four entries for 10.1.1.101, two with expiry 30/09/2026 and two Never (CK column redacted).*

C10 Existing policy

Menu path / command	SmartConsole > Security Policies > Access Control > Policy (tabs Standard and Alpha-Standard)
Observation	Policy inspected: Alpha-Standard, the policy shown for A-GW-Cluster (Figure A6g). Two policy tabs exist, Standard and Alpha-Standard, and the Install Policy dialog lists the same two (Figures C13b, C13c). Alpha-Standard has seven rules: 1 Noise rule; 2 Security Gateway Access rule; 3 Stealth rule; 4 Ping rule; 5 DNS rule; 6 Outbound Rule; 7 Cleanup rule (Figure S2a). Rule inspected: No. 1, Noise rule. Name Noise rule; Source Any; Destination Any; VPN Any; Services & Applications NBT and bootp; Action Drop; Track None; Install On shows Policy Targets (truncated). Summary pane: created by cpadmin, 5/23/2025 3:38 PM, expiration Never, hit count 469 (3%, Low), comment "Rule for clearing noise traffic from the network" (Figure S2b). Other rules as displayed: Security Gateway Access rule, Alpha-Admins to A-GW-Cluster, ssh and https, Accept, Log. Stealth rule, Any to A-GW-Cluster, Any, Drop, Log. Ping rule, Alpha-Admins to Alpha-Networks, icmp-requests, Accept, Log. DNS rule, Alpha-Networks to Any, dns, Accept, Log. Outbound Rule, A-SMS, A-GUI and

	A-MGMT-NET to Any, Any, Accept, Log. Cleanup rule, Any to Any, Any, Drop, Log. A separate single-rule view shows a Cleanup rule with Any / Any / Any / Any and Accept, created by System on 10/15/2024, hit count 405 (Figure S2c). The tab name is not visible in that capture, so I cannot say which policy it belongs to.
Interpretation	Read from the visible values, the Noise rule discards NetBIOS (NBT) and bootp traffic from any source to any destination, and its Track of None means these drops are not logged. Placed first, it removes broadcast chatter before the later rules can log it. The overall policy is default-deny: the Stealth rule protects the cluster, specific rules accept admin ssh and https, ping, DNS and outbound traffic, and the final Cleanup rule drops everything else with logging. The hit count shows traffic has matched the rule at some point. Nothing was edited.
Figure	S2a, S2b, S2c

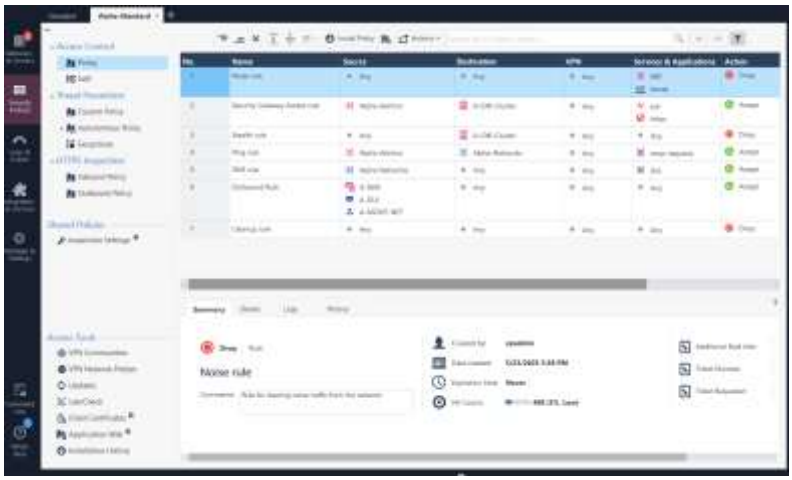


Figure S2a. *Security Policies > Access Control, Alpha-Standard tab: seven rules, with the Noise rule selected.*

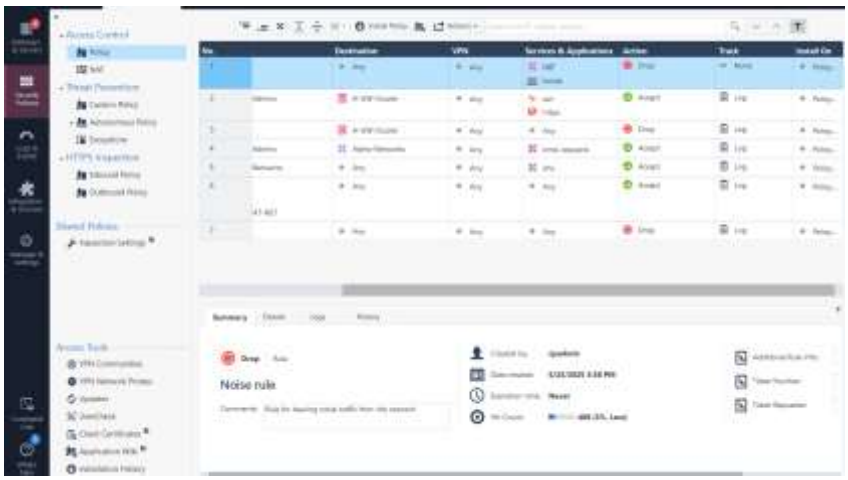


Figure S2b. *Same Alpha-Standard rule table scrolled right: Track and Install On columns, and the Noise rule Summary pane.*

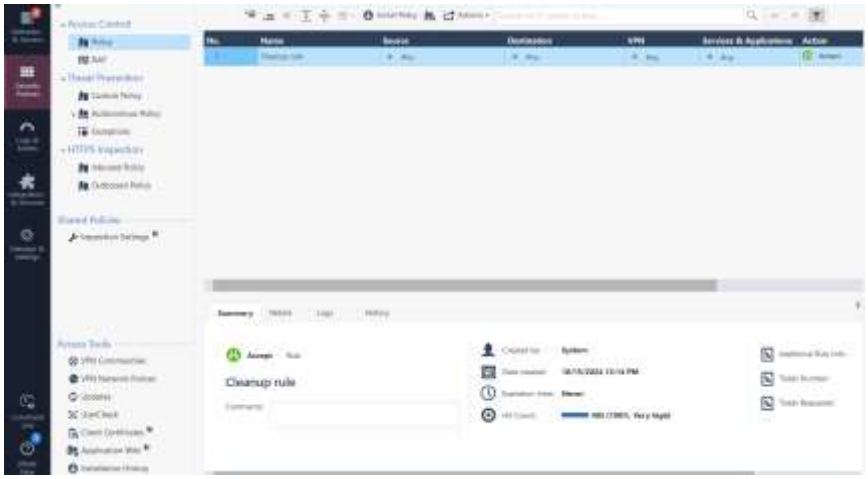


Figure S2c. *A second policy view with a single Cleanup rule (Accept); the policy tab name is not visible in this capture.*

C11 Objects and administrative controls

Menu path / command	SmartConsole > Objects pane (right) and Gateways & Servers; Manage & Settings; top toolbar; status bar
Observation	A-SMS: a Security Management Server (crown icon), address 10.1.1.101, comment "Alpha Site Primary Security Management Server", management blades Network Policy Management and Logging & Status; its purpose is to hold policy and logs. A-GW-Cluster: a gateway cluster, address 10.1.1.1, comment "Alpha-Site border gateway cluster", access blades Firewall and Monitoring, policy Alpha-Standard; its purpose is to enforce policy at the site border. Object categories: Network Objects 37, Services 522, Applications/Categories 8313, VPN Communities 2, Data Types 62, Users/Identities 1, Servers 1, Time Objects 3, UserCheck Interactions 15, Limit 4. Manage & Settings (left rail) opens Permissions & Administrators, Blades, Cloud Firewall, Sessions, Tags, Preferences, Sync with UserCenter, SmartTasks, Package Repository and Policy Settings (Figure C13). Session controls: Discard, Session, Changes and Publish at the top, all greyed with no pending changes, and the status bar reads No changes. Tasks window: bottom left, reading No tasks in progress.
Interpretation	Manage & Settings holds administrator accounts, blade and session administration. The session controls manage the working session: Discard abandons changes, Changes lists them and Publish commits them. The Tasks area shows long-running actions such as policy installation. Greyed controls and "No changes" are consistent with an unmodified session.
Figure	A6g (objects, session controls, Tasks), C13 (Manage & Settings)

C12 Logs and Events

Menu path / command	SmartConsole > Logs & Events > Logs
Observation	Time range Last 24 Hours, no query text, "Found 12 results (358 ms)". A yellow banner reads "Problems have occurred during search". Its details read "The following problems have been found: Log server is disconnected (Name: A-SME - 10.1.1.130)" (Figure S4c). All 12 entries have Origin A-SMS and run from Yesterday 11:38:11 PM to Today 9:54:06 AM. Source, Destination and Service are empty. The selected entry (Today 9:54:06 AM) shows Log Server Origin A-SMS (10.1.1.101), blade SmartEvent Client, product family Network, type Control, severity Medium.
Interpretation	The entries are control logs generated by A-SMS itself rather than traffic through a gateway. The warning shows that the separate log server A-SME was unreachable, which fits the lab design in which only A-SMS is running. The view therefore cannot show whether traffic or security events occurred elsewhere: the gateways are not deployed and one of the log sources was disconnected. No traffic was generated and no logging setting was changed to force a result.
Figure	S4a, S4b, S4c

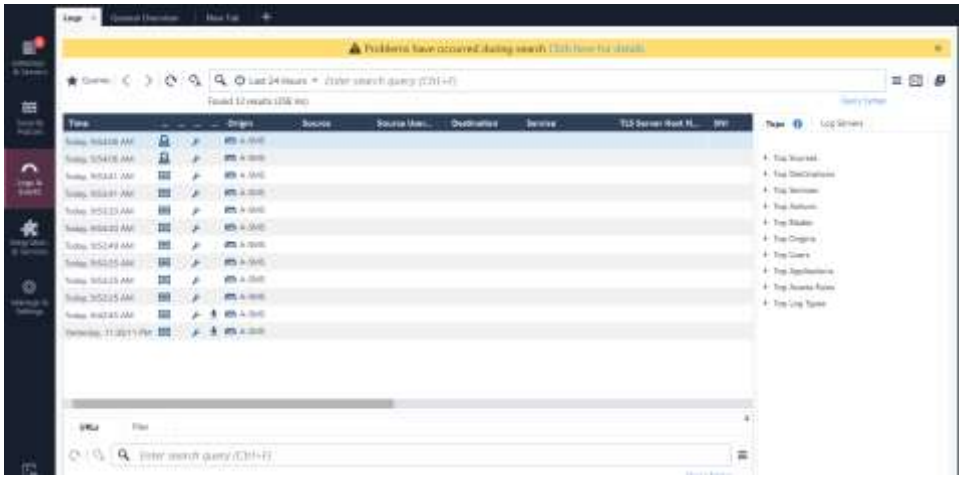


Figure S4a. *Logs & Events: Last 24 Hours, 12 results, all with origin A-SMS; a search-problem banner is displayed.*

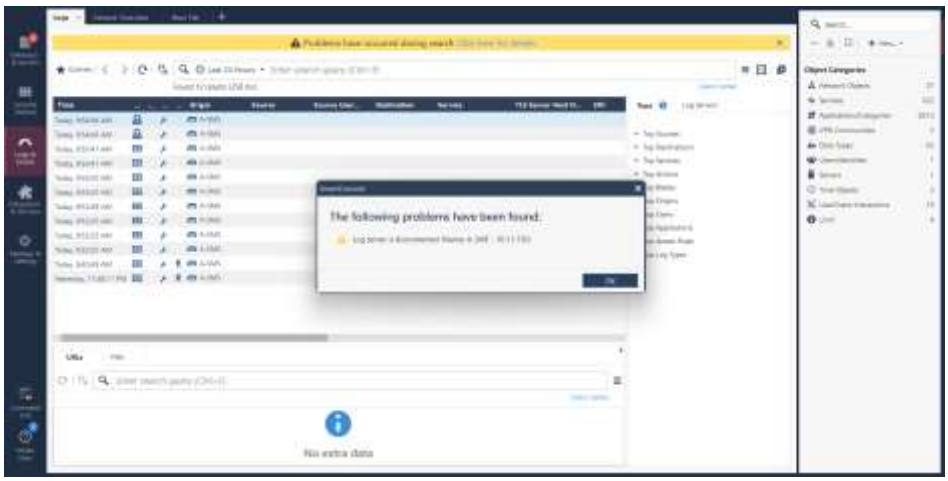


Figure S4c. *Search-problem details: Log server is disconnected (Name: A-SME - 10.1.1.130).*

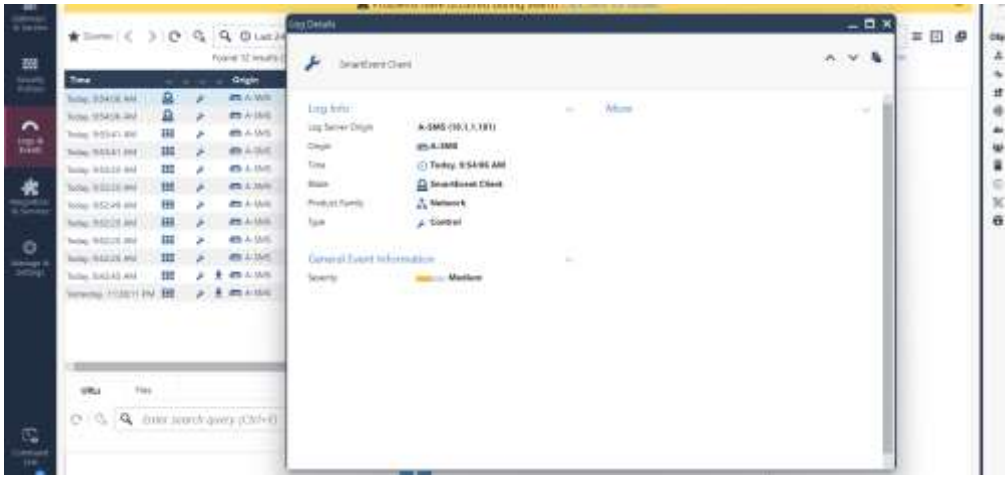


Figure S4b. *Log Details of the 9:54:06 AM entry: SmartEvent Client, type Control, severity Medium, origin A-SMS (10.1.1.101).*

C13 Publication and installation controls

Menu path / command	SmartConsole top toolbar and policy toolbar; Command Line (left rail)
Observation	Publish sits in the session group at the top (with Discard, Session and Changes) and is greyed. Install Policy is on the top toolbar and in the policy toolbar (Figures S2a, C13). The Install Policy dialog was opened to see its layout. Its Policy drop-down lists Standard and Alpha-Standard (Figure C13). With either selected, the dialog shows Access Control and Threat Prevention ticked, the message "No changes from last installation (4/12/2026)", the target A-GW-Cluster (10.1.1.1, R82, red status), an Install Mode section, and Install and Cancel buttons (Figures C13b, C13c). I clicked Cancel and did not install. The status bar reads No changes. Command Line is at the bottom of the left rail (Figures A6g, C8).

Interpretation	Publish saves the session's changes to the management database. Install Policy sends a published policy to gateways for enforcement (see B4). Neither was used. Slides 17 and 19 describe the installation and session controls. The message "No changes from last installation" shows the management server tracks what was last installed on a gateway separately from what is published. The Command Line panel is a management-command window, unlike Gaia Clish (OS configuration commands) and Expert mode (Linux Bash).
Figure	C13, C13b, C13c, A6g, C8

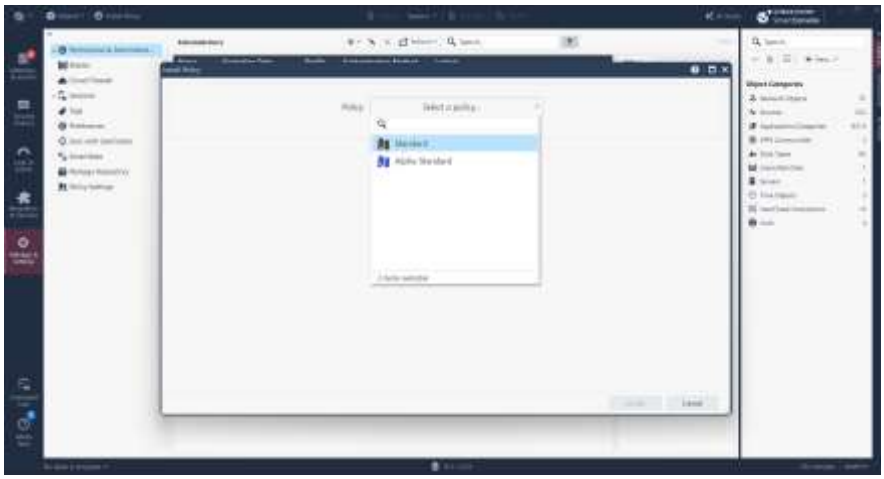


Figure C13. *Install Policy dialog opened for observation: policy list shows Standard and Alpha-Standard; Install button disabled; Manage & Settings visible behind it.*



Figure C13b. *Install Policy dialog with Standard selected: target A-GW-Cluster (red status), no changes from last installation (4/12/2026). Closed with Cancel.*



Figure C13c. *Install Policy dialog with Alpha-Standard selected: same target and message. Closed with Cancel.*

Issue Log

Part A entries record what the evidence shows. Part C findings were documented without making configuration changes.

Task	Symptom	Check	Observed result	Correction or next step	Fig.
A1	SmartConsole build differs between login screens (1027) and About window (1071)	SmartConsole About window	Both R82; updates enabled	Client auto-updated (confirmed by me). No action.	A1c, A6d
A2	Adapter summary showed NAT	VM settings, Network Adapter	NAT before; Custom VMnet8 after	Set to Custom VMnet8, Connect at power on; vmxnet3 line present	A2a, A2b, A2c
A5	eth0 verification	show interface eth0	Already 10.1.1.101/24, state on, link up	No correction needed	A5
A6	My first ping capture targeted 10.1.1.201	Compared target with the VM address	Own host answered (TTL 128)	Retested against 10.1.1.101: 4/4 replies, TTL 64	A6a
A6	Browser showed Not secure for the Gaia Portal	Confirmed the address is my own eth0	Self-signed certificate	Accepted warning for my own VM	A6c
C1	No CPU or memory on Gaia Overview	Scrolled the Overview page	No such panel in this build	Used SmartConsole values	G1, S1a
C3	No MAC or admin state in Portal interface table	Network Interfaces page	Columns absent	Used Clish values	G3, A5
C4	Default route 10.1.1.1 while NAT gateway is 10.1.1.250	IPv4 Static Routes, Virtual Network Editor	Values differ	Recorded, unchanged; internet not tested	G4a, A3b

Task	Symptom	Check	Observed result	Correction or next step	Fig.
C5	NTP not synchronised	System Management > Time	NTP servers not synchronized	Recorded, unchanged	G5a
C6	First show interfaces all capture lacked the eth0 section	Reviewed capture	Output scrolled past eth0	Re-ran and captured eth0 and lo sections	G6b, G6c
C7	Clish login showed CLINFR0771 config-lock message	Read message at login	Lock owned by admin	Did not use lock override; read-only commands unaffected	G7b
C8	ip route and ping -c rejected	SmartConsole Command Line	Syntax error and unsupported flag	Expected behaviour; no change	C8
C9	Two licence entries show 30/09/2026; Jumbo Hotfix messages differ	Licenses tab, banner, Gaia overview	Summary still OK	Recorded, unchanged	S1a, S1c
C12	Banner "Problems have occurred during search"	Opened banner details	Log server A-SME (10.1.1.130) disconnected	Expected, as only A-SMS is deployed; no traffic generated; no setting changed	S4a, S4c
C13	Install Policy dialog opened to observe its layout	Selected each policy to read the dialog	Target A-GW-Cluster red; no changes from last installation	Clicked Cancel; nothing installed	C13b, C13c

Conclusion

The lab shows the three-tier Check Point model in practice. SmartConsole on Windows administers the Security Management Server A-SMS over a private VMnet8 network, while enforcement belongs to a gateway that was only observed as a preconfigured object. Access was established in layers: network reachability by ping, the Gaia Portal over HTTPS, then a SmartConsole login that was trusted only after the ICA fingerprint on the VMware console matched. The Gaia and SmartConsole observations show a healthy management server and an unchanged, read-only session.

Limitations.

- The SmartConsole installer source is recorded as the instructor, and the download page was not captured.
- The Gaia Portal Overview has no CPU or memory panel in this build, so SmartConsole values are used.
- The tab name is not visible in the single-rule Cleanup (Accept) capture in Figure S2c, so that view is not tied to a policy.
- Object inspection used the Gateways & Servers summary pane; no object properties window was opened.
- Internet access from A-SMS was not tested, and the licence expiry dates, NTP state and route difference were not investigated.

References

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